UNIVERSITY OF CENTRAL PUNJAB LAHORE

Assignment # 4 Version 1

CLO Mapping: CLO3

Course Code	MAT243	Semester	Fall 2025
Course Title	Multivariable Calculus		
Resource Person	Dr. Muhammad Rayees Ahmad		
Assignment Given Date	25-12-2025	Submission Date	29-12-2025
Total Marks	20 = 15 (Assignment) + 5 (Quiz-based Viva)		

Version	Name of student for Version 1	Complete Registration no.	Section	Assignment marks	Viva marks	Total marks
1						

Submission Instructions *(Please follow strictly)*

- **Take the PRINTOUT OF THIS ASSIGNMENT PDF file and solve the questions on assignment papers or A4 papers. DO NOT use any other papers.**
- **This is an INDIVIDUAL ASSIGNMENT.**
- **Version 1 will solve the students who has the last digit ODD in their registration number.**
- **Version 2 will solve the students who has the last digit EVEN in their registration number.**
- **It is mandatory to know the whole method of all versions. Any question may be asked in the viva.**
- **Write your complete name and registration number on front page of the assignment.**
- **Follow the deadline. Finish your work one day before, so you could submit in time.**
- **LATE SUBMISSION WILL NOT BE ACCEPTED (Deduction of 2 marks each day).**
- **All Plagiarized assignments will be awarded ZERO (0) marks.**

I declare that I have prepared the assignment according to above guidelines, and I shall be responsible for any deduction of marks if the instructions are not followed.

Student signature: ______________________

Name: **Roll no.**

Q # 1 [CLO 3] (15 marks):

Use chain rule to find the derivatives of the function

$$h(x,y,z) = \ln(x^2 + y^2 + z^2)$$

where

$$x(r,s,t) = \cos(2rst)$$

$$y(r,s,t) = \sin\left(\frac{r}{s^2}\right)$$

$$z(r,s,t) = sin(2srt)$$

When $r = \frac{\pi}{4}\ , s = 2\ and\ t = -1$

Solution:

Name: **Roll no.**

Name: **Roll no.**

Name: **Roll no.**

Name: **Roll no.**

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Assignment # 4 Version 2

CLO Mapping: CLO3

Course Code	MAT243	Semester	Fall 2025
Course Title	Multivariable Calculus		
Resource Person	Dr. Muhammad Rayees Ahmad		
Assignment Given Date	25-12-2025	Submission Date	29-12-2025
Total Marks	20 = 15 (Assignment) + 5 (Quiz-based Viva)		

Version	Name of student for Version 2	Complete Registration no.	Section	Assignment marks	Viva marks	Total marks
2						

Submission Instructions *(Please follow strictly)*

- **Take the PRINTOUT OF THIS ASSIGNMENT PDF file and solve the questions on assignment papers or A4 papers. DO NOT use any other papers.**
- **This is an INDIVIDUAL ASSIGNMENT.**
- **Version 1 will solve the students who has the last digit ODD in their registration number.**
- **Version 2 will solve the students who has the last digit EVEN in their registration number.**
- **It is mandatory to know the whole method of all versions. Any question may be asked in the viva.**
- **Write your complete name and registration number on front page of the assignment.**
- **Follow the deadline. Finish your work one day before, so you could submit in time.**
- **LATE SUBMISSION WILL NOT BE ACCEPTED (Deduction of 5% marks each day).**
- **All Plagiarized assignments will be awarded ZERO (0) marks.**

I declare that I have prepared the assignment according to above guidelines, and I shall be responsible for any deduction of marks if the instructions are not followed.

Student signature: ______________________

Name: **Roll no.**

Q # 1 [CLO 3] (15 marks):

Consider a group of computer science students collaborating on a programming project.

Two students, Alice and Bob, are working on a complex algorithm together. The efficiency of the algorithm is crucial for the success of their project. They decide to measure the time complexity of their algorithm with respect to two variables: the size of the input data ($\boldsymbol{n}$) and the number of iterations in their loop ($\boldsymbol{m}$).

Consider the time complexity ($\boldsymbol{T}$) of the algorithm designed by Alice and Bob, given by the function

$$\boldsymbol{T(n, m) = \ln(2n + 1) + \sin(3m^2)}$$

Now, let $\boldsymbol{n}$ and $\boldsymbol{m}$ be functions of two variables $\boldsymbol{x}$ and $\boldsymbol{y}$

$$\boldsymbol{n(x, y) = e^{3x}\cos(5y)}$$

And,

$$\boldsymbol{m(x, y) = x^3 + 3y^2}$$

Here, $\boldsymbol{x}$ and $\boldsymbol{y}$ represent independent variables that influence the size of the input data ($\boldsymbol{n}$) and the number of iterations ($\boldsymbol{m}$), respectively.

Find $\frac{\partial T}{\partial x}$ and $\frac{\partial T}{\partial y}$ and interpret the meaning of $\frac{\partial T}{\partial x}$ and $\frac{\partial T}{\partial y}$ in the context of the algorithm's time complexity and the changing input data size ($\boldsymbol{n}$) and loop iterations ($\boldsymbol{m}$) with respect to the variables $\boldsymbol{x}$ and $\boldsymbol{y}$.

Solution:

Name: **Roll no.**

Name: **Roll no.**

Name: **Roll no.**